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CLINICAL INVESTIGATION

Long-Term Outcomes of Ocular and Visual Preservation After Carbon Ion Radiation Therapy for Choroidal Malignant Melanoma

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Purpose: This study aimed to evaluate the long-term results of carbon ion radiation therapy (CIRT) for choroidal malignant melanoma (CMM), especially regarding the preservation of the eye and visual acuity (VA).

Methods and Materials: A total of 250 patients with intraocularly localized CMM treated with CIRT between January 2003 and September 2021 were included. The dose prescription included 60 to 85 Gy/4 to 5 fr, with only 68 Gy/4 fr used from 2018 onward. The rotating gantry system with scanning beams was introduced in April 2018. Adverse events (AEs) were graded according to the Common Terminology Criteria for AEs (version 5.0.). For secondary glaucoma, tumor-related visual field defects were excluded from the evaluation. For VA, 245 patients with VA ≥ light perception (LP) were followed up. Effective VA (≥20/200, Snellen equivalent), counting fingers, and LP were used as indicators.

Results: The median age was 55 (15-86) years. The T categories 1, 2, 3, and 4 were observed in 16 (6.4%), 41 (16.4%), 189 (75.6%), and 4 (1.6%) patients, respectively. With a median follow-up of 72.5 months, the 5- and 8-year overall survival rates were 87.5% and 84.2%, respectively; the 5- and 8-year local control rates were 94.4% and 92.9%, respectively. At the last follow-up, 19 of 250 patients (7.6%) underwent enucleation, 15 caused by local recurrence and 4 caused by AEs. Secondary glaucoma grades 1, 2, and 3 to 4 were observed in 22 (8.8%), 49 (19.6%), and 5 (2.0%) of patients, respectively. At the last follow-up, ≥ effective VA, ≥ counting fingers, and ≥ LP were maintained in 80 (33%), 120 (49%), and 154 (63%) of patients, respectively. Preservation rate of ≥ LP vision at 5 and 8 years after CIRT was 65.7% and 55.3%, respectively.

Conclusions: CIRT for CMM is a promising treatment for both tumor control and preservation of the eye and VA.

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Background

Choroidal malignant melanoma (CMM) is a rare malignant tumor in Japan, with an incidence of approximately 0.6 cases per million people annually, whereas its incidence is several dozen times higher in the West.[1,2] Although enucleation has long been used to treat these tumors, there is an increasing demand for radiation therapy (RT) as a radical alternative treatment to preserve the eye and vision. In addition to brachytherapy (mainly iodine-125 episcleral plaque brachytherapy), which has been used for a relatively long time,[3,4] external beam RT modalities such as charged particle RT (CPT) and stereotactic RT (SRT) have been developed in recent years.[5,6] Among these, CPT using protons and carbon ions is a highly dose-concentrating irradiation method that exploits the Bragg peak.[7] Proton RT (PRT) has been used since about 1970, mainly in the West, and there are several large-scale reports of its clinical use. Carbon ion RT (CIRT) for CMM was developed around 1995 at QST Hospital (formerly Hospital of the National Institute of Radiological Sciences) in Japan and was initiated for clinical use in 2001.[8] There are still few reports on CIRT, but it has dose concentrations comparable with PRT and even better biological effects and, as with PRT, is expected to be effective against radioresistant tumors such as melanoma.[9]

The main reason for enucleation after RT is radiation-induced glaucoma, which can caused by angiogenesis or hemorrhage.[10,11] Enucleation rates vary depending on the radiation modality used, with reports ranging from 4% to 25%.[4,10-16] For CIRT, the postirradiation enucleation rate caused by neovascular glaucoma (NVG) was reported to be 5.5% in 2007,[17] and 7.2% at 5 years in a 2013 report.[18] Since then, treatment techniques and ophthalmic management of complications have continued to improve.

Visual preservation is difficult depending on the location of the tumor. Proximity to the optic nerve or the macula, or the larger size (thickness and basal diameter) of the tumor, will cause visual loss because of radiation macular edema or radiation optic neuropathy.[12,13,16] Several other causes, such as retinal detachment, ischemia, and glaucoma, also contribute to vision loss.[12,13,19] There are few reports on the details of adverse events (AEs) following CIRT for CMM, particularly regarding vision loss, and investigation of this would also serve the needs of patients.

In this study, we report the long-term outcome of patients with CMM treated with CIRT at our institution, focusing on ocular preservation and visual acuity (VA), in order to understand the role of this modality in the management of CMM and to provide guidance for further improvements in treatment.

Materials and Methods

Patients

Patients with CMM who underwent curative CIRT at our hospital between June 2003 and September 2021 were analyzed retrospectively. Patients who had been treated until May 2003 were excluded because of insufficient data on their visual function. Regarding patient eligibility for CIRT, there were no size limitations, but those with obvious distant metastases or extraocular extension were excluded. Prior to CIRT, all patients underwent a systematic ophthalmologic examination by an ocular oncologist and had a diagnosis of malignant melanoma. Staging examinations included whole-body contrast-enhanced computed tomography (CT, 0.5 mm slice for the orbit), orbital magnetic resonance imaging (MRI, 1 mm slice), positron emission tomography CT, and iodo-amphetamine single photon emission CT.[20] Contrast-enhanced liver MRI and/or liver ultrasound were also performed where possible to confirm eligibility. Informed consent was obtained from patients to allow their personal information to be used for research purposes. This retrospective study was reviewed and approved by the Institutional Ethics Committee on Human Clinical Research.

Treatment

Details of the treatment method have been reported in a previous study.[18] Prior to the treatment plan, 2 titanium markers per eye were sewn on the sclera. The procedure was performed by a skilled ophthalmologist. The patient was immobilized in a supine position on a head frame connected to a bite block with a camera to monitor the movement of the affected eye and a light-emitting diode light to fix gaze direction. Figure 1 shows the fixture used in our hospital.

The gross target volume was delineated on a thin-slice CT with 1-mm slice thickness fusing contrast-enhanced MRI. The clinical target volume was defined as the gross target volume itself, whereas the planned target volume was generated from the clinical target volume plus a safety margin of 1 to 2 mm because of positioning uncertainty. Treatment plans were designed to cover the planning target volume with at least 95% of the prescribed dose, tolerating

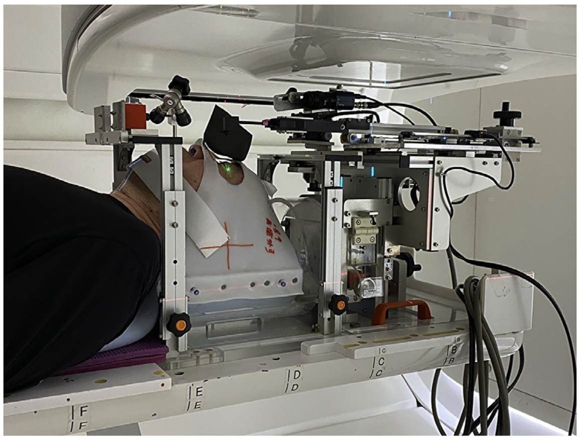

Fig. 1. A fixture used in our hospital. Patients were immobilized using a customized headrest, face mask, and bite block. A metal frame was also used to record coordinates and ensure reproducibility. The eyeballs were fixed using the gazing method.

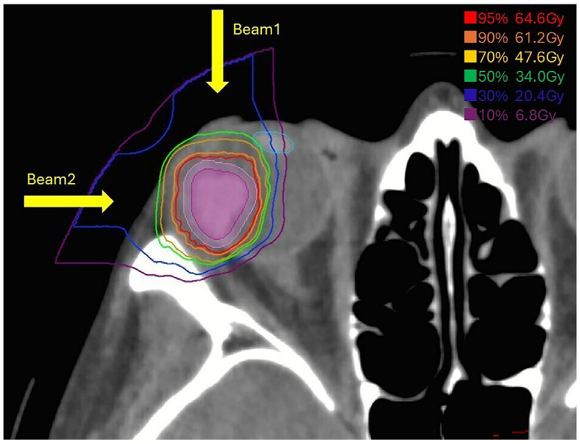

Fig. 2. A typical dose distribution of carbon ion radiation therapy for choroidal malignant melanoma. The red, orange, yellow, green, blue, and purple lines show doses of 95%, 90%, 70%, 50%, 30%, and 10%, respectively; gross target volume was drawn in magenta, and clinical target volume was in pale pink.

intratumoral hotspots (ie, areas irradiated more than 100%). At the beginning of the study period, our own treatment planning system EYEPLAN (NIRS) was used, but since April 2005 the planning system has been changed to the RT planning system XiO-carbon, FocalPro (CMS Inc). A typical irradiation field is shown in Figure 2. Prescribed doses were assessed using Gy (described as relative biological effect-weighted dose based on the modified microdosimetric kinetic model), calculated by multiplying the physical dose by the relative biological effect.[8,21] From the start of the treatment in 2001, a single vertical beam had been traditionally used, but in October 2005, it was changed to 2 beams, 1 vertical and 1 horizontal, to reduce the dose to organs at risk (OARs). Furthermore, from April 2018, rotating gantries and scanning technology have been adopted, allowing more controlled dose distribution. The prescribed dose started at 60 to 70 Gy/5 fr, followed by 64 Gy/4 fr (biologically effective dose [BED]10: 132-168 Gy) and the currently used prescription of 68 Gy/4 fr (BED10 183.6 Gy) from September 2016.

Follow-up and chart review

After CIRT, patients were followed up by both an ophthalmologist and a radiation oncologist. Whole-body CT, liver MRI or hepatic echography, and orbital MRI were performed every 3 months for the first 2 years and approximately every 6 months thereafter. Ophthalmologist consultations, VA, and intraocular pressure (IOP) monitoring were also routinely performed. AEs were updated at each visit, and if the exact onset or examination was unknown, the date of event confirmation was recorded as the onset. Acute and late AEs other than VA were assessed according to the Common Terminology Criteria for AEs Version 5.0,[22] with acute AEs defined as those occurring within 3 months and late AEs as those occurring thereafter. For the assessment of secondary glaucoma, preexisting visual field defects caused by the tumor or resulting retinal detachment were excluded. Other grading criteria were as follows: grade 1, asymptomatic and requiring no treatment; grade 2, requiring treatment with topical medications (including laser therapy) that reduced IOP to 21 mmHg or less; and grade 3 to 4, poorly controlled with medications or requiring more invasive procedures.[18] As indicators of VA, effective VA (EV, best corrected VA with Snellen chart of 20/200 or better), counting fingers (CFs), and light perception (LP) were used.

Local recurrence was defined as a contiguous increase in tumor proven by imaging examination or fundoscopy, or the development of a tumor adjacent to the primary lesion. Overall survival, progression-free survival, local control (LC), distant metastasis-free rates, and ocular and visual preservation were assessed from the date of initiation of CIRT to the date of the event or the date of the last follow-up and analyzed using the Kaplan-Meier method. Cox proportional hazards regression was used for the analysis of risk factors. Statistical analysis was performed using R software (https://www.r-project.org/) and the significance of the analysis was set at 2-sided $P < .05$.

Results

Patient characteristics

From June 2003 to September 2021, a total of 251 patients underwent CIRT, and all patients completed their course. Of these, 250 patients were included in the analysis, excluding 1 who was lost to follow-up without posttreatment imaging.

Patient characteristics are shown in Table 1. The median age was 55 (15-86) years; the T categories were 1, 2, 3, and 4 in 16 (6.4%), 41 (16.4%), 189 (75.6%), and 4 (1.6%) patients, respectively. The median tumor thickness was 7.5 (2-17) mm, and the median diameter of the tumor base was 12 (4-16.5) mm. Fifty-two (20.8%) tumors are attached to the optic disc, and 24 (9.6%) tumors invade beyond the ora serrata to the ciliary body.

Treatment effect

With a final follow-up of 31 September 2023, with a median follow-up of 72.5 (range, 5-236) months and 78 (range, 6-236) months for survivors only. Of the 250 patients, 44 (17.6%) died, of whom 36 were from the underlying disease and 8 from other causes. The 5-year and 8-year overall survival rates were 87.5% (82.2%-91.3%) and 84.2% (78.2%-88.6%), respectively, and the 5-year and 8-year progression-free survival rates were 77.9% (71.8%-82.8%) and 70.1% (62.9%-76.2%), respectively (Fig. 3A, B). Local recurrence

Table 1 Patient characteristics (n = 250)

Factors		
Follow-up period (mo)	Median (range)	72.5 (5-236)
Age (y)	Median (range)	55 (15-86)
Sex	Male:female	114:136
Tumor basis diameter (mm)	Median (range)	12.0 (4.0-16.5)
Tumor height (mm)	Median (range)	7.5 (2.0-17.0)
Size category	1:2:3:4	16:41:189:4
Tumor-disc distance	0 mm: >0 mm	52:198
Tumor-ciliary body distance	0 mm: >0 mm	24:226
Tumor-macula distance	0 mm: >0 mm	44:206
Dose prescription	60 Gy/5 fr	24
	70 Gy/5 fr	125
	64 Gy/4 fr	7
	68 Gy/4 fr	94

Abbreviation: fr = fraction.

was observed in 17 patients, and the LC rates at 5-year and 8-year were 94.4% (90.3%-96.8%), and 92.9% (88.1%-95.8%), respectively (Fig. 3C). Of the 17 cases of local recurrence, 15 had undergone enucleation, and 2 had undergone additional CIRT. Meanwhile, distant metastases were found in 66 patients, with a total of 71 first distant metastases: 53 in the liver, 11 in the lung, and 7 in other sites. The 5-year and 8-year distant metastasis-free rates were 79.8% (73.8%-84.5%) and 72.9% (65.8%-78.8%), respectively (Fig. 3D).

Secondary glaucoma and ocular preservation

Grade 2 acute AEs to skin and eye (excluding glaucoma and VA-related) occurred in 11 (4.4%) and 10 (4.0%) patients, respectively. No grade 3 events occurred. Regarding glaucoma, grades 1 or above were observed in 76 patients (30.4%), with grades 1, 2, and 3 to 4 glaucoma observed in 22 (8.8%), 49 (19.6%), and 5 (2.0%) patients, respectively. Of the 5 cases assessed as grades 3 to 4, 1 required enucleation. Other cases were treated with procedures, such as filtering surgery and vitreous surgery. Neovascularization was suspected in most cases of glaucoma, but in some cases, the effects of hemorrhage and infection could not be excluded, and neovascularization occurred because of a combination of causes.

Univariate analysis of predictors of the development of grade ≥2 glaucoma showed that tumor thickness (≥8 mm, P = .0055) was significantly associated, whereas tumor-OARs (optic disc and ciliary body) distance, number of ports (P = .61), and prescribed dose (P = .15) were not. Phthisis bulbi was also observed on long-term follow-up, and the radiographic diagnosis was confirmed in 20 patients (8%); 7 of the 20 patients were postglaucoma treatment. No clear trend was found in relation to tumor stage or location, but all cases were irradiated before 2014 and no occurrence in subsequent cases was observed.

Up to the final follow-up, 19 out of 250 patients (7.6%) underwent ocular enucleation.

The causes included local recurrence in 15 cases (79%) and AEs in the remaining 4 cases. Detailed causes of AEs included neovascular glaucoma, vitreous hemorrhage, uveitis, and infected corneal ulcers, each in 1 case. Enucleations were performed 29 (6-164) months after treatment, almost reflecting the timing of recurrence, whereas in the 4 patients with AEs, it was performed relatively early, between 6 and 28 months.

Preservation of VA

Of 250 patients, 245 patients who had ≥ LP were followed up for VA after CIRT: 226 (90%) had VA above CFs and 184 (74%) had EV. The median observation period for VA was 61 (3-236) months. At the last follow-up, ≥ EV (20/200), ≥ CFs, and ≥ LP were maintained in 80, 120, and 157 patients, respectively. Posttreatment LP loss (blindness) was observed in 88 patients (36%). The median time to blindness was 31.5 (2-204) months, and at 1, 2, 5, and 8 years after treatment, the maintenance of vision above LP was observed in 90.7% (86.3%-93.8%), 84.5% (79.1%-88.6%), 65.7% (58.4%-72.0%), and 55.3% (46.8%-62.9%) patients. Regarding the risk factors for loss of LP, tumor thickness was the most significantly correlated factor in both univariate and multivariate analyses (>8 mm, P = .00000050, hazard ratio [HR], 3.77 {2.25-6.32}), and tumor-ciliary distance (P = .039, HR, 1.90 [1.03-3.48]) was also significantly associated. No significant association was observed between increased dose and loss of LP (P = .59). (Table 2)

Although detailed longitudinal data on EV were not available, the same factors as for LP were examined for EV loss at the last observation; tumor thickness (>8 mm, P = .00024, HR, 3.71 [1.84-7.46]) as for LP, and tumor-macular distance (0 mm distance, P = .015, HR, 3.75 [1.29-10.9]) were significantly associated. Tumor-ciliary distance showed a negative correlation in contrast to LP (0 mm distance, P = .035; HR, 0.33 [0.12-0.93]). A higher dose prescription (BED10 183.6 Gy) also tended to be associated with rather better VA preservation (P = .013, HR, 0.44 [0.23-0.84]).

Discussion

Significance of RT

RT is now the most commonly used treatment as it provides comparable LC while preserving the eye and VA.[23] Although

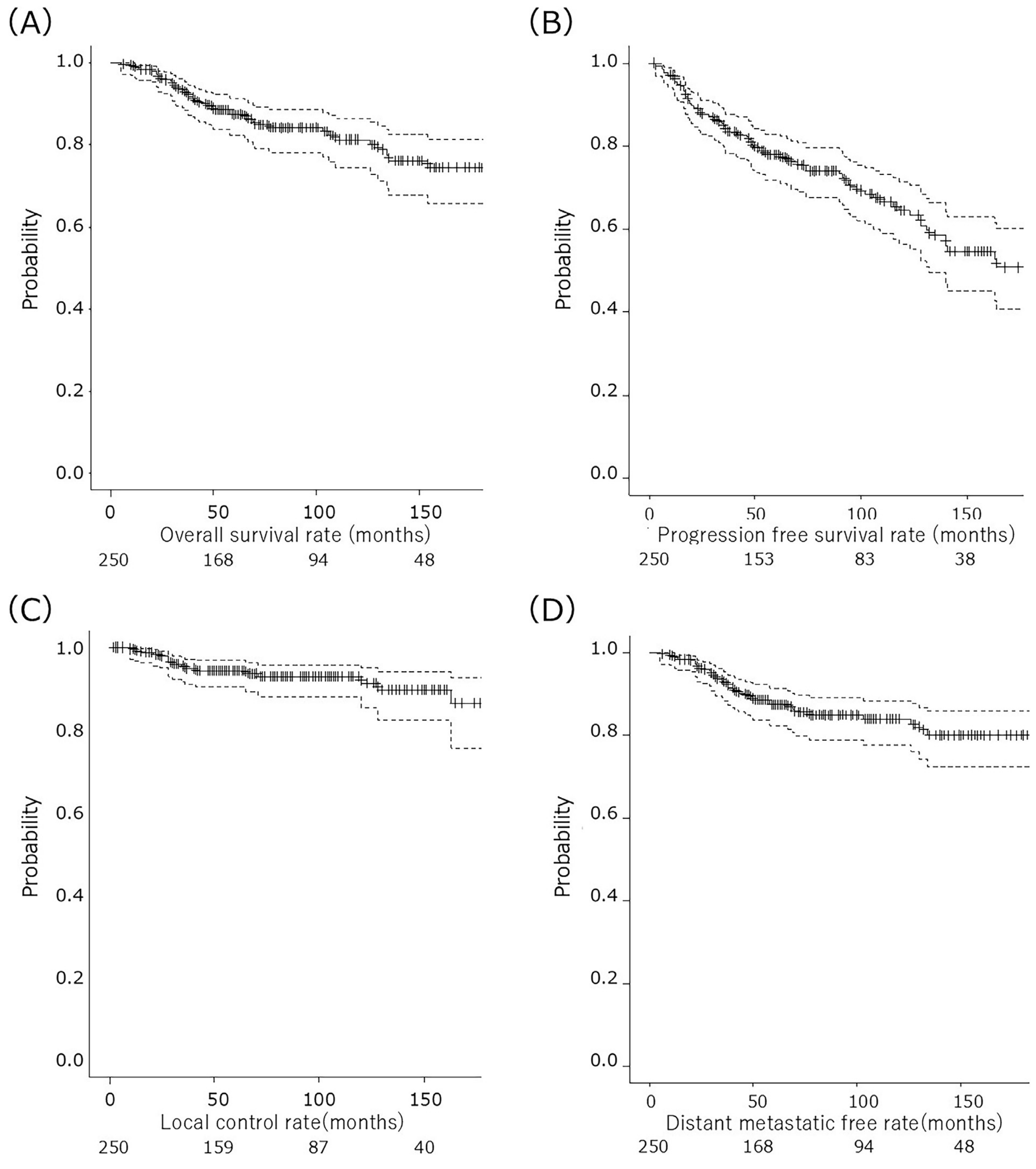

Fig. 3. Kaplan-Meier curves of (A) overall survival, (B) progression-free survival, (C) local control, and (D) distant metastasis-free rates in the whole cohort (n = 250).

brachytherapy is indicated to treat CMM, including T4, difficulties have been reported with the placement of seeds for tumors in proximity to the disc and for bulky tumors,[3] which is supported by several clinical reports.[4,15,24] Consistent with the United Kingdom national audit reports,[24] CPT (mainly CIRT) has been performed in Japan for CMMs that are too large for brachytherapy or are posterior poles. The results of a prospective, randomized trial comparing brachytherapy and CIRT reported in 2015 showed that CPT significantly improved LC, ocular preservation, and disease-free survival.[25] Among the CPT, PRT is a well-established RT modality for CMM, supported by several large reports.[14,15,26-28] In contrast, reports on CIRT are still scarce, and this is the largest to our knowledge.

Reports on RT for CMM are summarized in Table 3.[4,13-16,26-31] The present study showed survival rate and LC comparable with those observed in the previous reports at a sufficiently long observation period (median, 72.5 months) despite relatively large tumor sizes.[4,14,27,31,32] Also, most previous reports in external irradiation using high doses, equivalent

Table 2 The patient and treatment factors associated with loss of light perception

Evaluation factors	Indicators	Univariate *P* value	Multivariate *P* value	HR (95% CI)
Age (y)	(>50 vs ≤50)	.13	.33	
Sex	(Male vs female)	.020*	.50	
Diameter of tumor	(>12 mm vs ≤12 mm)	.0010*	.24	
Thickness of tumor	(>8 mm vs ≤8 mm)	.0000000047*	.00000050*	3.77 (2.25-6.32)
Tumor-disc distance	(0 mm vs >0 mm)	.050*	.19	
Tumor-ciliary body distance	(0 mm vs >0 mm)	.85	.41	
Tumor-macula distance	(0 mm vs >0 mm)	.057	.039*	1.90 (1.03-3.48)
Dose prescription (BED10)	(132-168 Gy vs 183.6 Gy)	.60	.59	
No. of ports	(1 vs 2)	.86	.58	

Abbreviations: BED = biological effective dose; HR = hazard ratio.
* $P = <0.05$.

to 60 to 70 Gy/4 to 5 fr, have also achieved LC of greater than 90%, regardless of the RT modality, suggesting adequate dose delivery for CMM.[13,26-28,31] The higher incidence of distant metastases, especially liver metastases, compared with that of local recurrence was similar to that observed in previous reports. A significant increase in the dose prescription is not compatible with eye and VA preservation and could be considered a limitation of focal treatment. Investigation of concomitant drug therapies and their indication criteria is an important issue for future studies.[33]

Eye preservation and IOP control

In the present study, the eye preservation rate (EPR) after CIRT was 93.2% at 5 years and still >90% at 8 years. This compares favorably with the reported 5-year EPR of 82.0% to 95.6% after brachytherapy,[4,29,34] 74% to 84% 5-years after SRT,[10-12,30] 75.3% to 93.0% after PRT in the previous reports.[13,15,27,28,31] Of the 19 (7.6%) enucleations, 4 (1.6%) were caused by AEs, which is an improvement of 2.6% in our previous report published in 2013.[17] Looking at other modalities, Eibenberger et al[16] found that 54 out of 335 patients (16%) required enucleation after SRT for CMM, including 41 (12%) for NVG. In a report on PRT, Bergman et al[32] reported that 77 of 579 patients required enucleation within 3 years of irradiation, of which 17 (2.9%) were caused by AEs. While Sagoo et al[4] reported that 27 of 141 cases (19%) underwent enucleation, of which 20 (14.2%) were caused by AEs. Compared with these reported results, the main reason for the favorable EPR in the present study was maybe because of the lower incidence of AEs, especially

Table 3 Summary of previous reports on radiation therapy for choroidal malignant melanoma

Author	Year	No.	Modality	Dose/fraction	Follow-up (mo)	Height (mm)	Diameter (mm)	5y-OSR (%)	5y-LCR (%)	5y-DMFR (%)	5y-EPR (%)
Sagoo et al[4]	2014	650	Brachy	80 Gy	40	3.5	10	96	86	89	84
Rouberol et al[29]	2004	213	Brachy	-	56.5	-	-	82.0	78.3	-	82.0
Egger et al[26]	2001	2435	Proton	60-70 CGE/4 fr	40	5.8	16	-	95.8	-	-
Egger et al[14]	2003	2645			44	5.8	16	-	-	-	88.9
Dendale et al[27]	2006	1406	Proton	60 CGE/4 fr	73	4.8	13	79	96	80.6	93
Caujolle et al[28]	2010	886	Proton	60 CGE/4 fr	63.7	5	15.7	79.4	93.9	88.3	91.1
Damato et al[15]	2005	349	Proton	53.1 CGE/4 fr	37	3	10.1	-	96.5	-	90.6
Fuss et al[13]	2001	78	Proton	70.2 CGE/5 fr	34	6	10	70.3	90.5	76.2	75.3
Tran et al[31]	2012	59	Proton	54-60 CGE/4 fr	63	3.5	11.4	85	91.3	82.1	80
Al-Wassia et al[30]	2011	50	SRT	60 Gy/10 fr	29	4	12	94	85	91	84
Eibenberger et al[16]	2021	335	SRT	50-70 Gy/5 fr	78.6	4.67	-	85	96.7	84.7	82.7
Current	2024	250	CIRT	60-70 Gy/5 fr, 64-68Gy/ 4fr	72.5	7.5	12	87.5	94.4	79.8	93.2

Abbreviations: CGE = Cobalt Gray equivalent; CIRT = carbon ion radiation therapy; DMFR = distant metastasis-free rate; EPR = eye preservation rate; fr = fraction; LCR = local control rate; OSR = overall survival rate; SRT = stereotactic radiation therapy.

enucleation for severe glaucoma. This can be caused by several technical reasons, including advances in irradiation techniques and improved AE management.[17,35,36] Detorakis et al[37] identified neovascularization of the anterior chamber or retina as a possible cause of NVG development. Indeed, Riechardt et al[38] reported that ciliary body dose was associated with the development of grade ≥2 glaucomas. Irradiated posterior segment volume and optic nerve length were reported to be significantly associated with NVG,[17,18,39] and Fernandes et al[40] reported that pathologic changes and dose distribution in patients undergoing enucleation showed that NVG was mainly caused by radiation damage in the posterior rather than the anterior segment. In the present study, multivariate analysis of predictors of glaucoma grade ≥2 showed that none of the tumor-OAR distances were significantly associated, but only tumor thickness. Similarly, tumor size (thickness or basal diameter) has been identified in previous reports as an indirect factor for increased dose to normal intraocular structures.

Regarding the timing of enucleation, Fuss et al[13] reported that enucleation because of AEs was required within a short period of 15 months. In the present study, the 2 earliest enucleations at 6 and 8 months were because of glaucoma and vitreous hemorrhage, respectively. However, unpredictable AEs, such as infection, have also led to enucleation >2 years after irradiation.

Preservation of VA

Few reports have described visual loss and its course after RT for CMM, especially not for CIRT; Fuss et al[13] reported that in 57 patients who underwent PRT, posttreatment VA was effective (Snellen equivalent, ≥20/200) in 49% of patients, but the observation period in this study was rather short, with a median of 34 months and a maximum of 102 months. Eibenberger et al[16] found that the proportion of patients who maintained best corrected VA with Snellen charts of 20 per 200 or better after SRT was 56.8% at 1 year, 5.4% at 5 years, and 0.6% at 10 years. In the present study, 33% of patients preserved VA ≥ EV at the last follow-up over a relatively long observation period with a median of 61 months, which is considered satisfactory.

Detailed data on the time course of EV, which is important for patients' quality of life, were not available in this study and were instead examined using LP. The results showed that light loss occurred gradually without a peak, up to 10 years after CIRT. This is the same trend as observed by Eibenberger et al[16] who followed the long-term course of VA after PRT, although the treatment and assessment methods were different. Previous reports have also shown that, as with IOP, multiple OAR doses may be involved in postirradiation loss of VA,[12,41] which may indicate a gradual and complex course in the absence of a single determinant factor. In the present study, tumor thickness and tumor-ciliary distance (0 mm) were associated with LP loss. Similar factors have been reported in previous studies[42] and are also common factors in the development of NVG. It has been reported that tumor thickness influences the total dose to intraocular structures and correlates with VA. In contrast, when focusing on EV preservation at the last observation, tumor thickness and tumor-macular distance (0 mm) were significantly associated, whereas tumor-to-ciliary distance (0 mm) was a more inverse factor. This is probably caused by the strong influence of radiation-induced optic neuropathy on EV, which is dose-dependent in the posterior fovea. Fuss et al[13] also reported that the visual prognosis was dependent on the dose (>35 CGE) to key structures in the posterior chamber, ie, the macula and disc, and the OAR-tumor distance can be used as an alternative predictor of the OAR dose. In contrast, no significant associations were found for tumor-disc distance. The stronger finding for the ciliary body than for the disc may be caused by the different tolerable doses of these 2 organs.[43] Further research is needed to clarify the tolerable dose for each structure by analyzing the dose-volume histogram. The better preservation of vision in the high dose group, ie, the current regimen (68 Gy in 4 fractions), seems to be caused by the relatively short observation period. This is also a possible result of improved treatment methods, and, at least for the time being, our higher doses do not seem to have a negative effect on vision.

Limitations

This study has several limitations. First, it was a single-center, retrospective study. Because of the long follow-up period, patient backgrounds and details of treatment methods may have varied. Dose-volume histogram data were not available for some cases in the first half of the study and could not be directly checked for AEs and OAR dose thresholds. In addition, the older patients included in the study may not have been adequately analyzed because of incomplete follow-up and lack of detailed clinical data. Second, the limited number of patients may have reduced the statistical power of the results. Also, there are no established methods for comparing the biological effects of CIRT with other modalities, making accurate comparisons with previous reports of other modalities difficult. Third, because we specialize in CIRT, ophthalmologic examinations, and treatment are conducted by ophthalmologists at other institutions. Therefore, it was not possible to obtain all the information on the time course of IOP and VA, as well as on the diagnosis using fundoscopy and ultrasound. In the diagnosis of glaucoma, the assessment of visual field loss caused by tumors or radiation is difficult and could be underestimated. In addition, VA was recorded in stages using loss of 20 per 200, CF, and LP as indicators, so a detailed time course could not be assessed continuously. AEs were not always assessed on the day of onset but at the next visit to the hospital. Also, the methods and indications for the treatment of AEs may have varied between institutions.

Conclusions

This study reported the long-term results of CIRT for CMM. The data showed that CIRT is an effective and safe treatment for CMM. Although the preservation of the eyes and their function was favorable, the loss of VA occurred over a long period, and there is room for improvement. Management of AEs requires close collaboration with ophthalmologists. To maximize the benefits of CIRT in preserving the eyes and VA, more detailed data collection and analysis of risk factors, as well as further development of treatment techniques based on the data, are required.

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